Prodigy Anesthesia Quick Review

Serotonin Antagonists

What is the most common use for serotonin antagonists?	They are most commonly used as prophylactic antiemetics. Depending upon the specific serotonin receptor they antagonize, however, serotonin antagonists can be used to treat nausea and vomiting, carcinoid syndrome, gastroesophageal reflux, and irritable bowel syndrome. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1278-1279.*
What are the side effects of 5-HT3 receptor antagonists?	5-HT3 receptor antagonists exhibit no serious side effects, even in large doses. The most common side effects are headache and a slight prolongation of the QT interval in some circumstances. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 285.*
Are the serotonin receptors coupled to G-proteins or are they ion channels?	All of the serotonin receptors are G-protein-coupled and affect either adenylyl cyclase or phospholipase C with the exception of the 5-HT3 receptor which is an ion channel. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 284.*
How is serotonin metabolized?	Serotonin is metabolized by monoamine oxidase into 5-hydroxyindoleacetic acid *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 282.*
What is the principal use for the serotonin antagonist, cyproheptadine?	Cyproheptadine is a 5-HT2 receptor antagonist that also has Histamine-1 receptor antagonist properties. It is used to treat the smooth muscle dysfunction associated with carcinoid syndrome. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 286.*
What is the prototypical 5-HT3 serotonin antagonist?	Ondansetron *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 287.*
How do 5-HT3 receptor antagonists prevent nausea?	5-HT3 receptor antagonists exert their antiemetic effects by blocking 5-HT3 receptors centrally in the vomiting and chemoreceptor trigger zone as well as peripherally in the intestinal vagal and spinal afferent nerves. *Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 1098.*

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What is the dose of ondansetron in pediatric patients for the treatment and prophylaxis of postoperative vomiting?	0.1 mg/kg *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1192.*
How is ondansetron metabolized?	Ondansetron undergoes extensive metabolism by the liver via hydroxylation and conjugation by the cytochrome P-450 pathway. As a result, the elimination half-life is prolonged in patients with liver failure and the dosage may need to be reduced. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 285.*

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